

CBSE Test Paper 02
Chapter 10 Mechanical Properties of Fluids

1. Which one of the following statements is correct for a fluid passing through the narrow part of non-uniform pipe? **1**
 - a. its velocity decreases but its pressure increases
 - b. its velocity and pressure both increase
 - c. its velocity increases but its pressure decreases
 - d. its velocity and pressure both decrease
2. Flow is said to be steady if **1**
 - a. the flow rate decreases 10% every second
 - b. the flow rate does not change with time
 - c. the flow rate increases with time
 - d. the flow rate decreases 20% every second
3. Gauge pressure at a point is **1**
 - a. pressure at that point – atmospheric pressure
 - b. pressure at that point + atmospheric pressure
 - c. (pressure at that point – atmospheric pressure)/2
 - d. (pressure at that point + atmospheric pressure)/2
4. Water drops are spherical because of **1**
 - a. elasticity
 - b. viscosity
 - c. low shear strain
 - d. surface tension
5. The unit of surface tension in S.I units is given by **1**
 - a. dynes per cm
 - b. dynes per cm²

-
- c. Newtons per meter²
 - d. Newtons per meter

6. Hydrostatic pressure is a scalar quantity even though pressure is force divided by area, and force is a vector. Explain? **1**
 7. Does Archimedes principle hold in a vessel in a free fall? **1**
 8. The blood pressure in human is greater at the feet than at the brain. Why? **1**
 9. What do you mean by velocity head, gravitational potential(elevation) head and pressure head? Write Bernoulli's equation in terms of pressure head, velocity head and elevation head. **2**
 10. A body of mass 6 kg is floating in a liquid with $\frac{2}{3}$ of its volume inside the liquid. Find ratio between the density of the body and density of liquid. Take $g = 10\text{m/s}^2$. **2**
 11. A liquid drop of diameter D breaks up into 27 tiny drops. Find the resulting change in energy. Take surface tension of liquid as σ . **2**
 12. If eight rain drops each of radius 1 mm are falling through air at a terminal velocity of 5 cm | s. If they coalesce to form a bigger drop, what is the terminal velocity of bigger drop? **3**
 13. If a drop of liquid breaks into smaller droplets, it results in lowering of temperature of the droplets. Let a drop of radius R, break into N small droplets each of radius r. Estimate the drop in temperature. **3**
 14. What is Bernoulli's theorem? Show that sum of pressure, potential and kinetic energy in the streamline flow is constant? **3**
 15. Briefly explain the role of detergent in washing of clothes. **5**
-